
Contents

1	Introduction	2
2	Origin of Acoustic Body Signals	3
2.1	Cardiac and Vascular Sounds	3
2.2	Respiratory Sounds	3
2.3	Gastrointestinal Sounds	3
2.4	Joints and Tendons	3
3	Sensor Types	4
3.1	Low Frequency Sensors	4
3.1.1	Seismographic Sensors	4
3.2	High Frequency Sensors	4
3.2.1	Piezoelectric Sensors	4
3.3	Ferroelectret Sensors	4
4	Discussion	5
5	conclusion	6

1 Introduction

The human body emits acoustic signals that can provide valuable insights into a vast array of physiological and pathological processes. Ranging from the rhythmic beating of the heart to subtle sounds produced by the respiratory system and gastrointestinal tract or high frequency hits in joints, these signals offer a non-invasive window into the body's inner workings. Auscultation as one of the oldest diagnostic techniques, has been used since the days of Hippocrates (c.460 - c.370 BC) to assess the health of patients by listening to their internal sounds [1]. With the advent of modern sensor technology and advanced signal processing techniques the analysis of acoustic body signals has exceeded far beyond the capabilities of the traditional stethoscope and limited human hearing. This work explores the origins of various acoustic body signals and their respective spectrums. Providing excerpts of the physiological processes that generate these sounds and how they can be used for diagnostic analysis. In the second part of this work, an overview of in use and emerging sensor technologies for acoustic emissions is given, focusing on their working principles and frequency ranges. Lastly, the detectability of acoustically emitted signals using 3D-printed ferroelectret sensors is discussed and compared to existing measurement approaches.

2 Origin of Acoustic Body Signals

2.1 Cardiac and Vascular Sounds

In a regular cardiac cycle the human heart emits sounds during two dominant events named the First Heart Sound (S1) and Second Heart Sound (S2)[2]. S1 is generated by the closure of the mitral and tricuspid valves at the onset of systole, while S2 stems from the halting of the aortic and pulmonary valves at the end of systole. In time-domain, S1 corresponds to the QRS complex of the electrocardiogram (ECG) and S2 to the T-wave. Using Phonocardiography (PCG) the acoustic signals of the heart can be recorded and analyzed for diagnostic purposes. A normal PCG signal has a frequency content in the range of 40 Hz to 200 Hz [3]. S1 emits .

2.2 Respiratory Sounds

2.3 Gastrointestinal Sounds

2.4 Joints and Tendons

3 Sensor Types

3.1 Low Frequency Sensors

3.1.1 Seismographic Sensors

3.2 High Frequency Sensors

3.2.1 Piezoelectric Sensors

3.3 Ferroelectret Sensors



4 Discussion



5 conclusion

Bibliography

- [1] M. R. Montinari and S. Minelli, "The first 200 years of cardiac auscultation and future perspectives," *Journal of Multidisciplinary Healthcare*, vol. 12, pp. 183–189, Mar. 6, 2019, issn: 1178-2390. doi: 10.2147/JMDH.S193904 Accessed: Apr. 17, 2026. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6408918/>
- [2] S. M. Debbal and F. Bereksi-Reguig, "Computerized heart sounds analysis," *Computers in Biology and Medicine*, vol. 38, no. 2, pp. 263–280, Feb. 1, 2008, issn: 0010-4825. doi: 10.1016/j.combiomed.2007.09.006 Accessed: Apr. 14, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0010482507001771>
- [3] S. Li, F. Li, S. Tang, and W. Xiong, "A review of computer-aided heart sound detection techniques," *BioMed Research International*, vol. 2020, no. 1, p. 5846191, 2020, _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1155/2020/5846191>, issn: 2314-6141. doi: 10.1155/2020/5846191 Accessed: Apr. 14, 2026. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1155/2020/5846191>